

Build night 2

Data Bars and Change Alerts

Bars from activity, not the clock. An agent that wakes only when something changes.

Travis Sim

NUS · Singapore

Thu 29 Oct 2026 · COM3/COM4

Where we are

Tonight's question: how many agent calls does a day of price ticks really need?

DONE

Build night 1

Thu 22 Oct

Data collection and the risk hook

TONIGHT

Build night 2

Thu 29 Oct

Data bars and change alerts

NEXT

Workshop 2

Thu 5 Nov

Labeling trades and forming teams

Tonight

18:00 to 20:00, mostly hands on.

18:00 — 10 min
Progress circle
last result and tonight's question

18:10 — 10 min
Mini-talk
clock bars, dollar bars, the CUSUM filter

18:20 — 20 min
Demo tour
you predict, then we run it

18:40 — 20 min
Guided mission
run it, predict, change, chart

19:00 — 50 min
Free build
your own project

19:50 — 10 min
Show and tell
push, one line on next step

Clock versus activity

Two ways to cut a stream of trades into bars.

CLOCK BARS

1 bar per minute



A quiet minute still makes a bar. A busy minute is squeezed into one.

ACTIVITY BARS

1 bar per $X traded



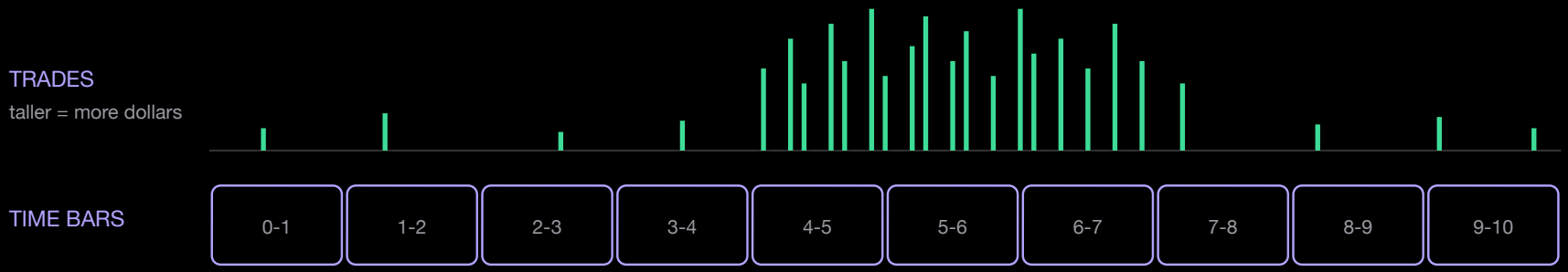
A quiet hour makes few bars. A busy minute makes many.

Illustrative

Time bars

Same trades, cut by the clock.

One bar every minute, whatever happened in it.



Quiet minutes hold one trade. The busy stretch is squeezed into a few bars.

Illustrative trades, $3,000 per dollar bar.

Dollar bars

Same trades, cut by money traded.

A bar closes when $X has traded, not when a minute has passed.



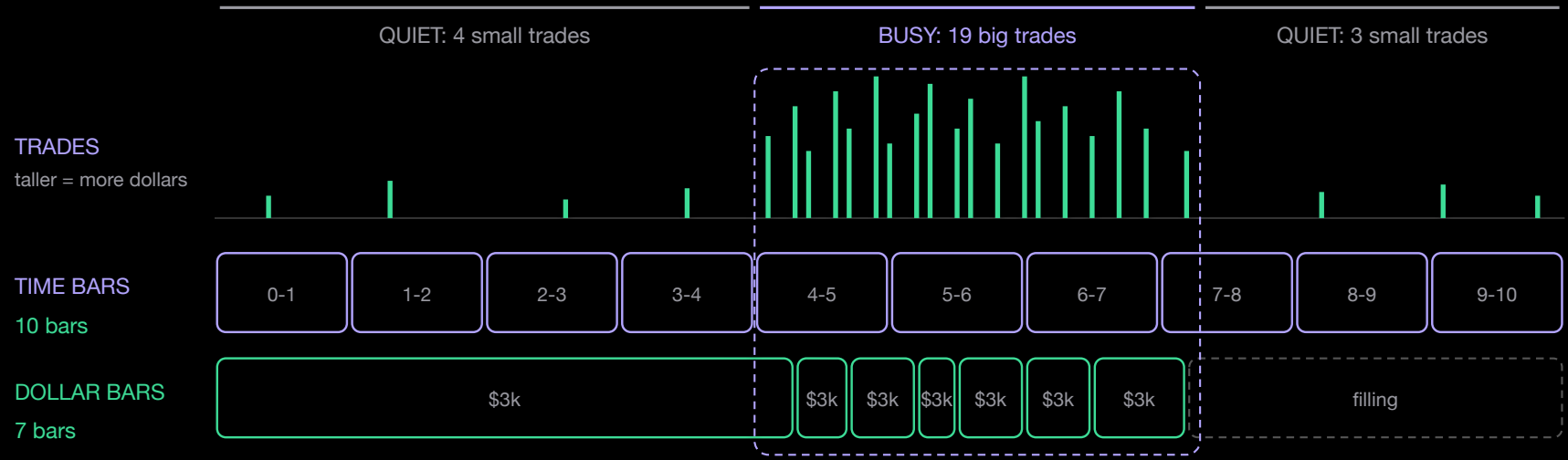
Quiet periods give few bars, busy periods give many.

Illustrative trades, $3,000 per dollar bar.

Same trades, both ways

Count the bars in the busy stretch.

The same 26 trades, cut two ways.



Busy stretch: 4 time bars, 7 dollar bars

Same trades. The dollar bars put their detail where the trading was.

Illustrative trades, $3,000 per dollar bar.

Changes add up

A change filter keeps a running total of price moves.



STEP 1 OF 2

Each tick moves the price a little. Add the moves up. Most ticks cancel out.

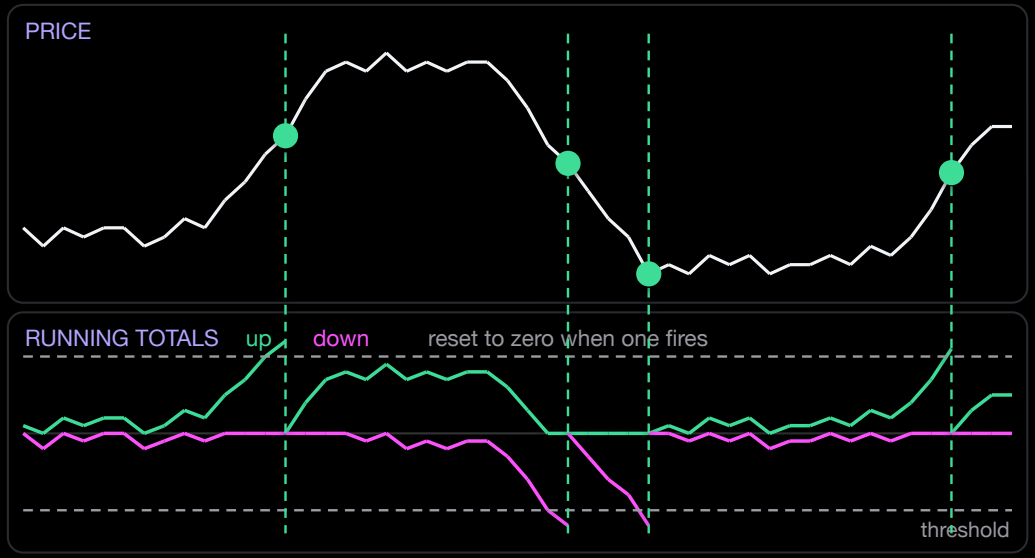
IN THE CODE

```
sPos, sNeg
```

Illustrative price moves.

The bucket fires

Cross the threshold, fire an event, empty the bucket.



STEP 2 OF 2

Up moves fill one bucket, down moves fill the other. Nothing happens until one is full.

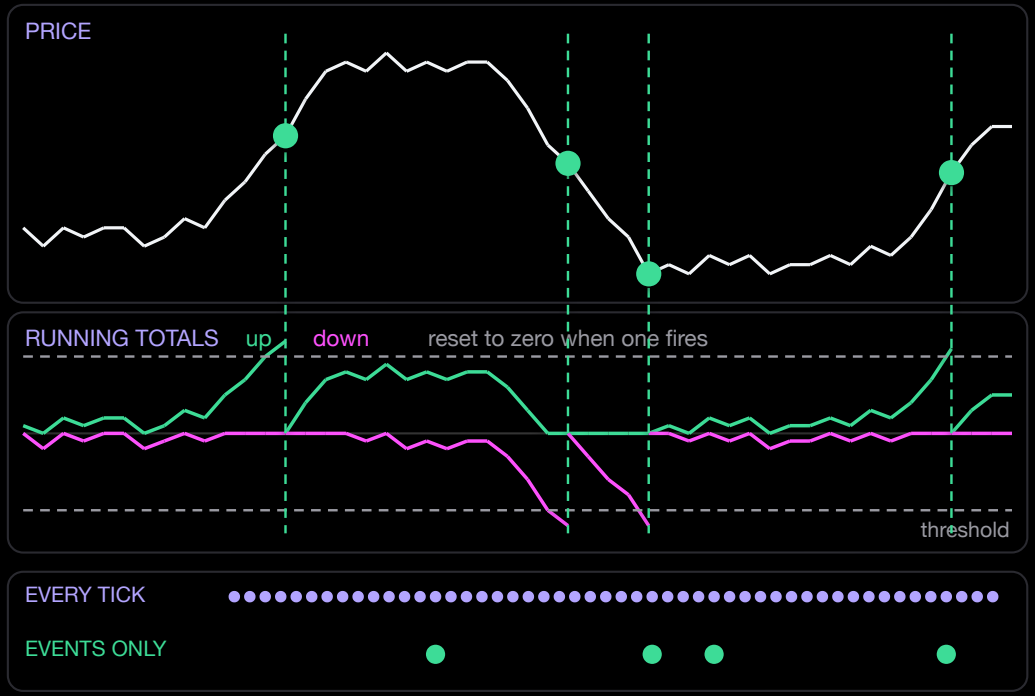
IN THE CODE

```
threshold
```

Illustrative price moves.

The agent wakes only on events

Every call to an agent costs money and time.



AGENT CALLS: EVERY TICK

681

wakes on every price tick

AGENT CALLS: EVENTS ONLY

24

wakes at the threshold (15 bps)

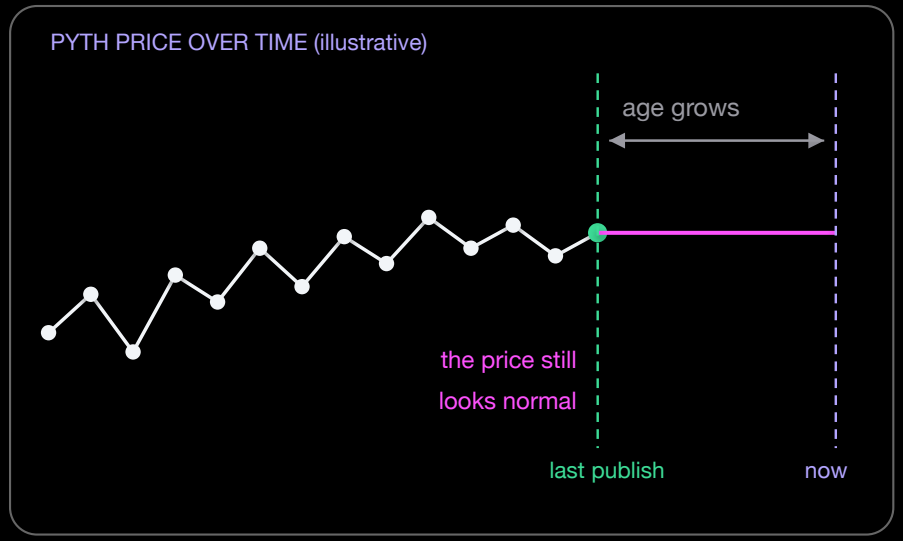
96% fewer calls

Simulated, on recorded data; not advice.

Illustrative chart; counts from demo-repo/code/out/cusum.json.

A frozen feed

Pyth carries a publish time and a slot. Compare both with now.



AGE IN SECONDS

`now minus publish time`

limit 30s

typical now: 11s

SLOTS BEHIND

`current slot minus posted slot`

limit 75 slots

typical now: 37 slots

Either one over its limit: stale. Do not act on it.

Illustrative chart; limits and typical values from demo-repo/code/out/staleness.json.

Real numbers

Real Orca SOL/USDC swaps, built into bars.

REAL SWAPS

2,759

Orca SOL/USDC pool

TRADED

$2,322,236

USDC side, same swaps

TIME BARS

43

60 s each, 0 empty

DOLLAR BARS

44

$50,000 each

WINDOW OF REAL SWAPS

2026-10-21 17:48 to 18:30 UTC

= 01:48 to 02:30 Singapore time (42 minutes)

Recorded data; a bar here is a summary of swaps.

Reads use public mainnet data; every write is devnet only; no real funds.

Source: code/out/swaps.json and bars.json, recorded data captured 2026-10-21 19:03:24 UTC. Orca SOL/USDC whirlpool swaps, public mainnet RPC.

A bar builder is short

Walk the swaps, add up dollars, close a bar at the threshold.

src/bars.ts

```
export function dollarBars(swaps: Swap[], thresholdUsd: number): Bar[] {
  if (!(thresholdUsd > 0)) throw new Error("dollar threshold must be > 0");
  const bars: Bar[] = [];
  let cur: Bar | null = null;
  for (const s of swaps) {
    if (!cur) {
      cur = startBar(s, s.t);
      bars.push(cur);
    } else addToBar(cur, s);
    if (cur.volumeUsd >= thresholdUsd) cur = null;
  }
  return bars;
}
```

1 **A plain loop**

Walk the swaps in order, add up the dollars.

2 **One number**

The threshold sets how big a bar is.

3 **Nothing lost**

Every swap lands in exactly one bar.

Source: demo-repo/code/src/bars.ts (excerpt: comments and blank lines removed)

Demo: time bars and dollar bars

PREDICT FIRST

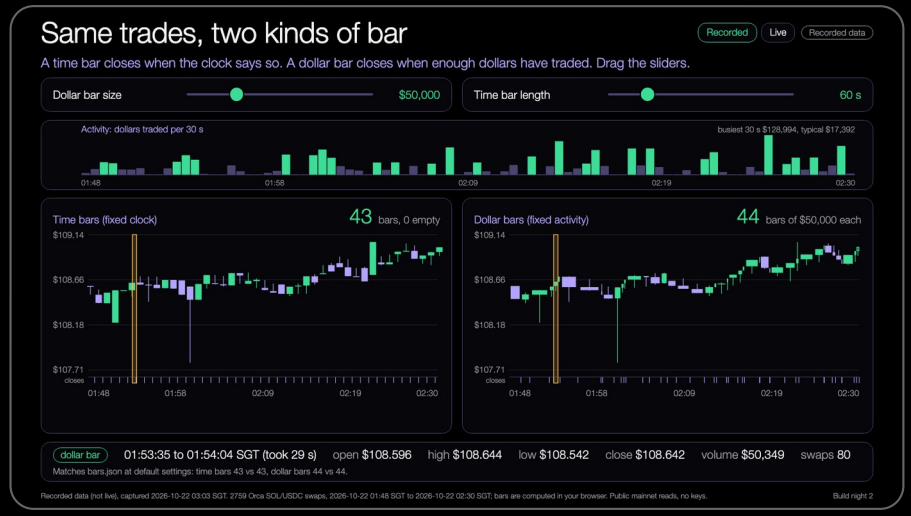
Which makes more bars from the same swaps, the clock or the dollars? By how many?

```
$ npm run demo -- --recorded
```

YOU SHOULD SEE

- bar counts for both kinds, from the same swaps
- one-minute bars that are empty

Fallback: Recorded data is the default. No network needed.



Learning Lab demo, runs from disk, recorded data.

Try it: https://nus-solana-demos.vercel.app/0003/01-time-vs-dollar-bars.html

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Demo: a change alert

PREDICT FIRST

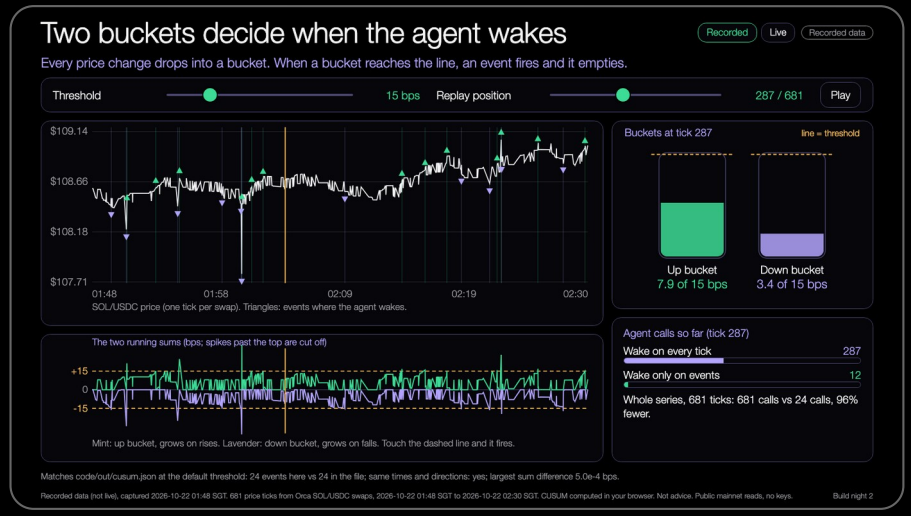
How many agent calls does the filter save, and where does it fire?

```
$ npm run demo -- --recorded --cusum-bps 15
```

YOU SHOULD SEE

- events marked up or down on the price
- agent calls: every tick versus events only

Fallback: Same recorded data. The events are the same every run.



Learning Lab demo, runs from disk, recorded data.

Try it: https://nus-solana-demos.vercel.app/0003/02-cusum-bucket.html

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Demo: a frozen feed

PREDICT FIRST

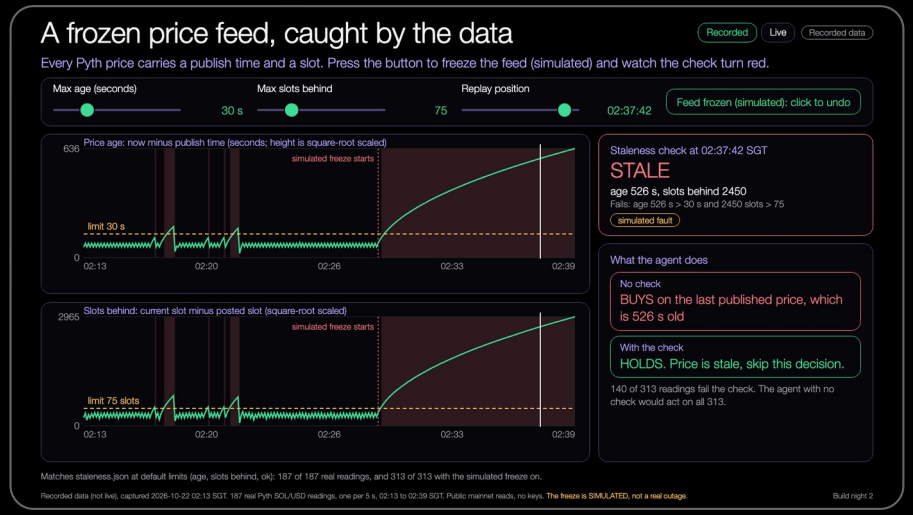
How many seconds until the staleness check turns red?

```
$ npm run demo -- --recorded --max-age-sec 30
```

YOU SHOULD SEE

- age and slots behind, over time
- the check turns red when a limit is crossed

Fallback: Same recorded data. The fault is simulated on top of it.



Learning Lab demo, runs from disk, recorded data.

Try it: https://nus-solana-demos.vercel.app/0003/03-frozen-feed.html

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Try the demos yourself

DEMOS

`nus-solana-demos.vercel.app/0003/`

Open on your laptop. Move the sliders.

No install. The same files are in the repo and open from disk.



Scan to open the demos

Source: demo-repo/lab (the four Learning Lab demos, deployed as static pages, recorded data)

Your mission

18:40 to 19:00. Twenty minutes.

1

Run

The sample bar builder

2

Predict

The bar count before you look

3

Change

The threshold or the token

4

Chart

The result, before and after

Finish line: one chart with a before and an after.

Build menu

19:00 to 19:50. Pick one, or bring your own project.

BEGINNER	Bars for another token
STANDARD	A CUSUM alert bot
STANDARD	A staleness monitor
ADVANCED	Wire bars into an earlier strategy and compare agent calls

Three takeaways

1. Activity bars carry more even information than clock bars.
2. A CUSUM filter wakes the agent only on a real change, so it makes fewer calls.
3. Check how old a price is, in seconds and in slots.

Branch out

After tonight, in your own time.

BEGINNER

Run the bar builder on another token, or halve the threshold, and compare the bar counts.

STANDARD

Try three CUSUM thresholds and chart agent calls against the events you catch.

ADVANCED

Feed your bars into an earlier strategy and compare agent calls.

Share back

Before you leave, and by next Thursday.

TONIGHT

1

Push your branch to your own repo.

THU 5 NOV

2

Bring your chart to Workshop 2.

ANY TIME

3

Send your repo link to the leads.

Get the code

REPO

`github.com/travisim/solana-agentic-trading-nus`

`folder: 0003-data-bars-change-alerts/demo`

```
1  $ npm install
```

```
2  $ npm run demo -- --recorded
```

Run both from the demo/code folder. Node 24. No keys, no AI credits. Recorded data is bundled.



Scan for this session's code

Let's connect



LinkedIn

linkedin.com/in/travisim



Telegram

t.me/travisim


SOLANA